DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B. Tech **Branch: Common to all Branches** **Semester: I**

Subject Code & Name: 24AF1000ES106 & Programming for Problem Solving

Max Marks: 60 **Date: 13/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	Which generation of computers introduced the use of integrated circuits (ICs)?				Remember	1
	a. First Generation	b. Second Generation	c. Third Generation	d. Fourth Generation		
2	Which of the following is a primary function of an operating system?				Understand	1
	a. Word processing	Understand	c. Image editing	d. File encryption		
3	Which component acts as the brain of a computer system?				Understand	1
	a. Input device	b. Processor	c. Output device	d. Memory		
4	Which of the following is not a valid C data type?				Remember	1
	a. int	b. char	c. bool	d. string		
5	What is the output of the following expression in C? int x = 10, y = 5; printf ("%d", x > y && y < 10);				Apply	1
	a. 0	b. 1	c. 5	d. 10		
6	Which operator is used for bitwise AND operation in C?				Remember	1
	a. &&	b. &	c. |	d. ||		
7	What is the correct syntax for a do-while loop in C?				Understand	1
	a. do {...} while(condition);	b. do { ... } while(condition)	c. while(condition) { ... } do;	d. do { while(condition); }		
8	Which statement is used to exit a loop prematurely in C?				Remember	1
	a. exit	b. break	c. continue	d. return		

9	What will be the output of the following code?
	int x = 5;
	if (x == 5)
	  printf("Hello");
	else
	  printf("World");
					Apply	1

	a. Hello	b. World	c. HelloWorld	d. Compilation Error		
10	What is the index of the first element in a C array?				Remember	1
	a. 1	b. 0	c. -1	d. Depends on the		

				array		
11	What does the following pointer declaration mean? int *ptr;				Understand	**1**
	a. ptr is a pointer to an integer	b. ptr is an integer	c. ptr is a pointer to a float	d. None of the above		
12	Which of the following is used to write data to a file in C?				Understand	**1**
	a. fread	b. fwrite	c. fprintf	d. All of the above		
Q. 2	**Solve the following.**					**12**
A)	What are the steps involved in programming? Briefly describe each step.				Understanding	**6**
B)	What is the role of memory management in a computer system? Differentiate between primary and secondary memory.				Analyze	**6**
Q.3	**Solve the following.**					**12**
A)	What is operator precedence? Write an expression and explain how it is evaluated.				Understand & Apply	**6**
B)	Describe the conditional (ternary) operator and write a program to find the maximum of two numbers using it.				Understand & Apply	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	Differentiate between while, for, and do-while loops with example programs.				Understand & Apply	**6**
B)	Write a c program to demonstrate the use of break and continue in loops.				Apply	**6**
C)	Describe the basics of user-defined functions and write a function to calculate the factorial of a number.				Understand & Apply	**6**
Q.5	**Solve Any Two of the following.**					**12**
A)	Explain the initialization of arrays in C with examples.				Understand & Apply	**6**
B)	Write a program to create and display a 3x3 matrix using a two-dimensional array in C.				Apply	**6**
C)	Write a c program to demonstrate the use of a pointer to an array.				Understand & Apply	**6**
Q. 6	**Solve Any Two of the following.**					**12**
A)	What is an array of structures? Write a c program to store and display details of 5 students.				Understand & Apply	**6**
B)	Discuss file opening and closing in c with examples of different modes.				Understand & Apply	**6**
C)	Write a c program to read and write data to a file using fprintf() and fscanf().				Understand & Apply	**6**
	***** End *****					